



Configural and Featural Processing in the Face Network

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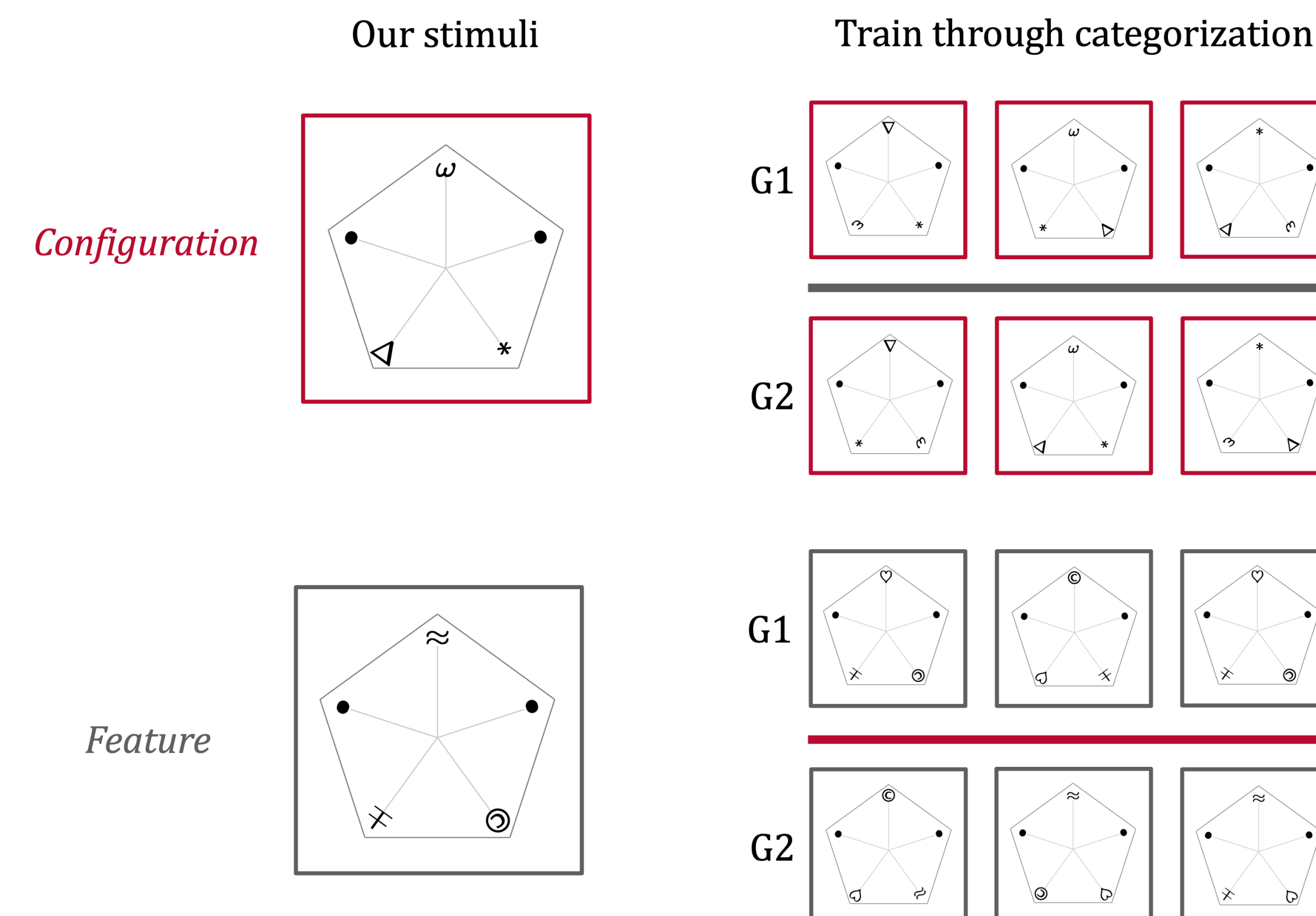
Background

Traditionally, *configural processing* has been regarded as primarily face-specific, essential for holistic face recognition [1,2], while *feature processing* is emphasized in both face and non-face objects recognition, focusing on independent features [3]. This dichotomy implies a specialized role for configural processing in face perception and a broader role for feature processing in general object recognition.

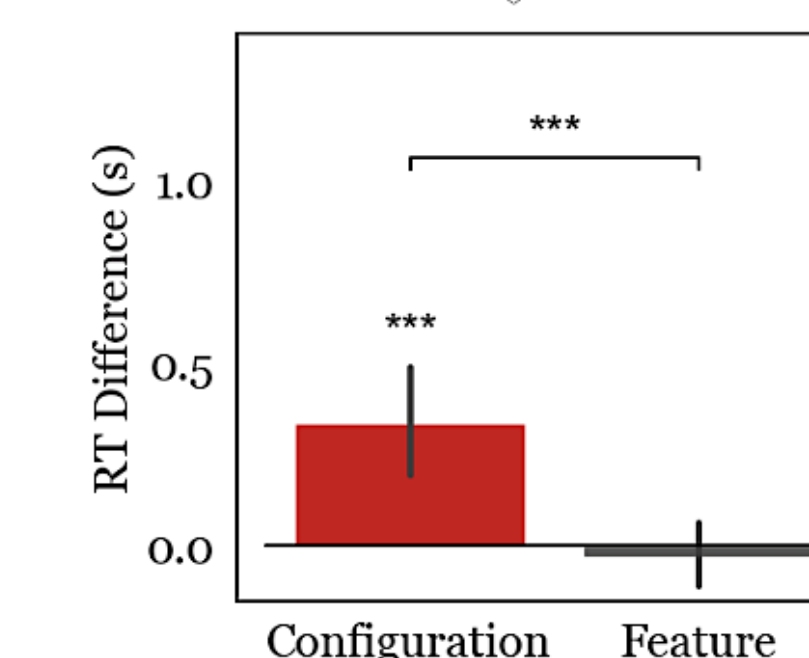
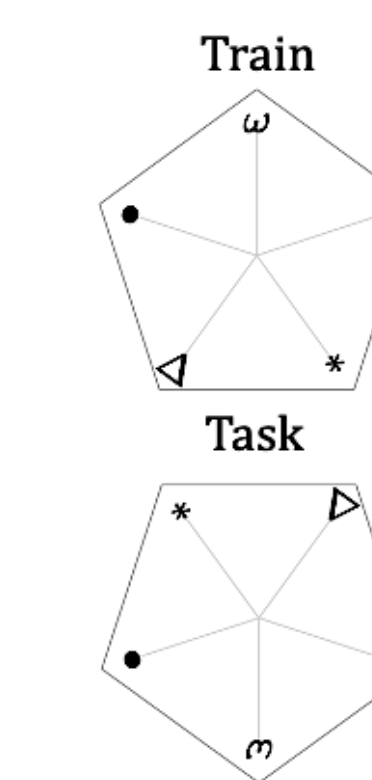
Our recent research challenges this view, suggesting that configural processing may be a fundamental, mid-level visual mechanism, not exclusive to faces. Last year's findings demonstrated that non-face objects that are categorized by configural properties exhibit all of the same phenomena as faces.

Building on this, our current study uses the same stimuli alongside feature-based controls to further investigate the neural mechanisms underlying configural versus feature processing. This approach aims to refine our understanding of how these processes function across different stimuli, potentially redefining their roles in visual cognition.

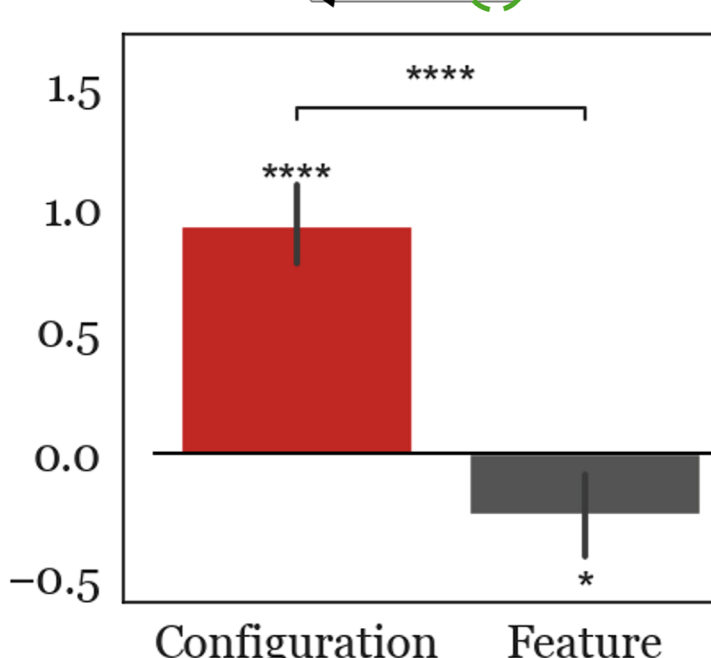
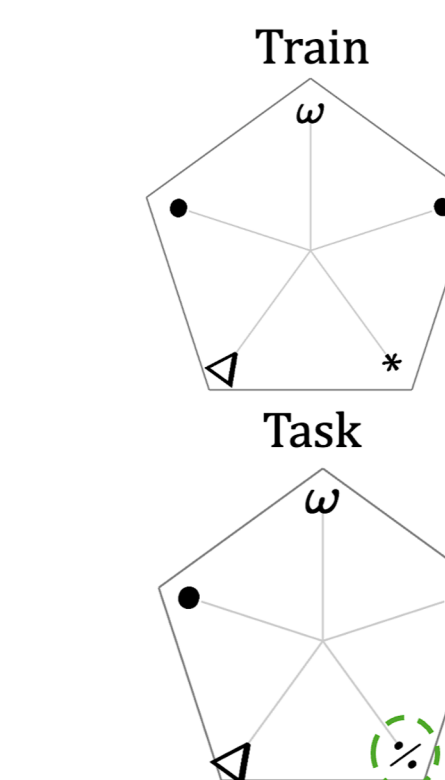
Our Previous Results



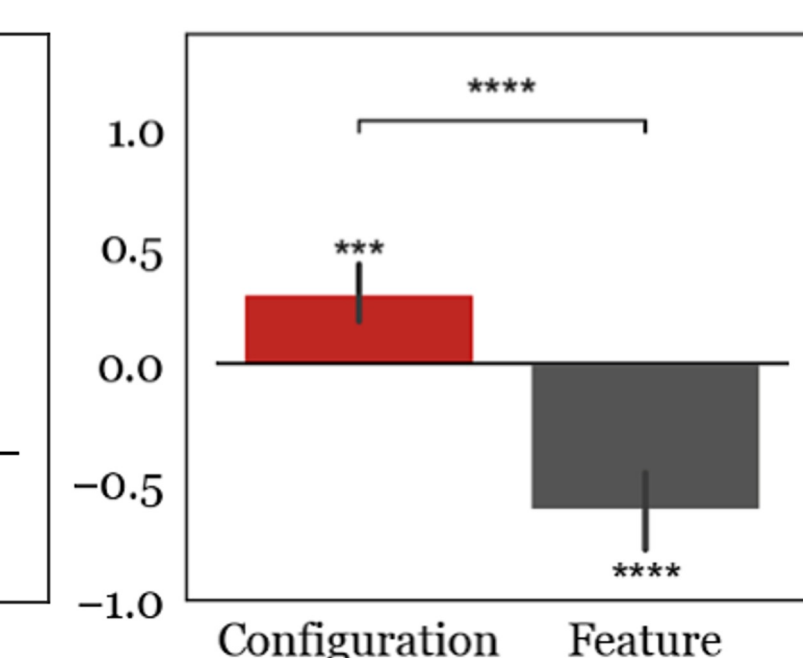
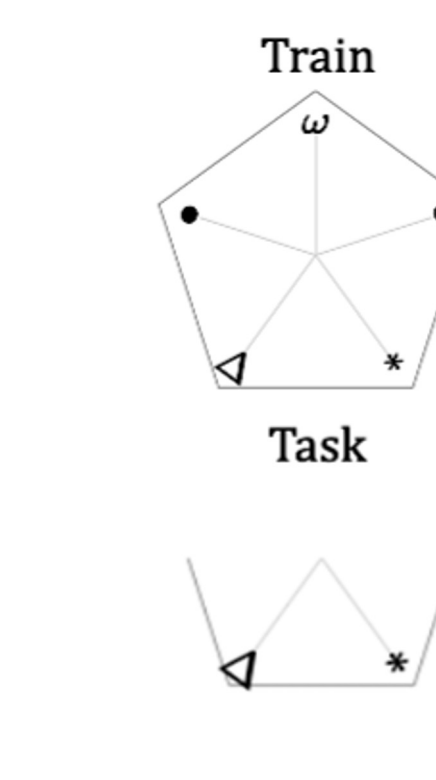
Inversion



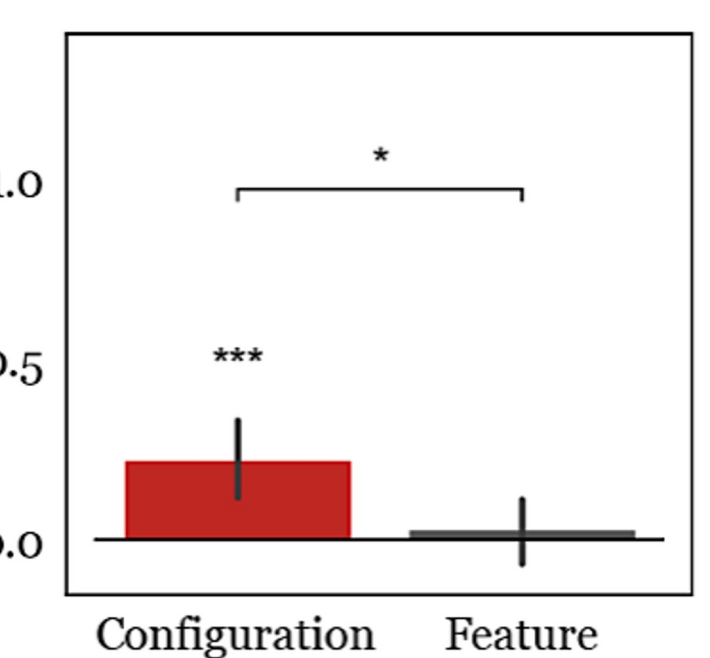
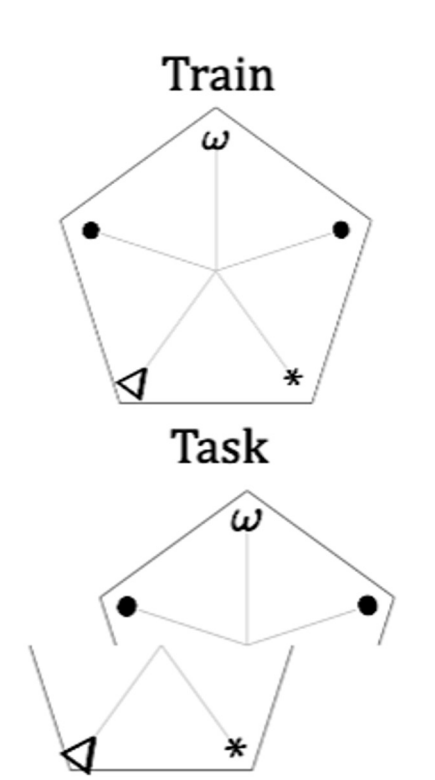
Composite



Part-Whole



Misalign



We also found these results in **human accuracy** and even **CCN models**. Check our last year's poster here:



Methods

To reveal the brain mechanism of configural versus feature processing, we scanned subjects after they learned the configural and feature information from our lab-made stimuli.

N=12 healthy adults (F=8).

Subjects were trained to categorize stimuli for **~1.5 hours** immediately before MRI session.

MRI sequence:

- Structural MRI
- Visual functional localizer [4]
- Configural-Feature task
 - Categorization task
 - Passive viewing task
- Resting state fMRI
- DTI

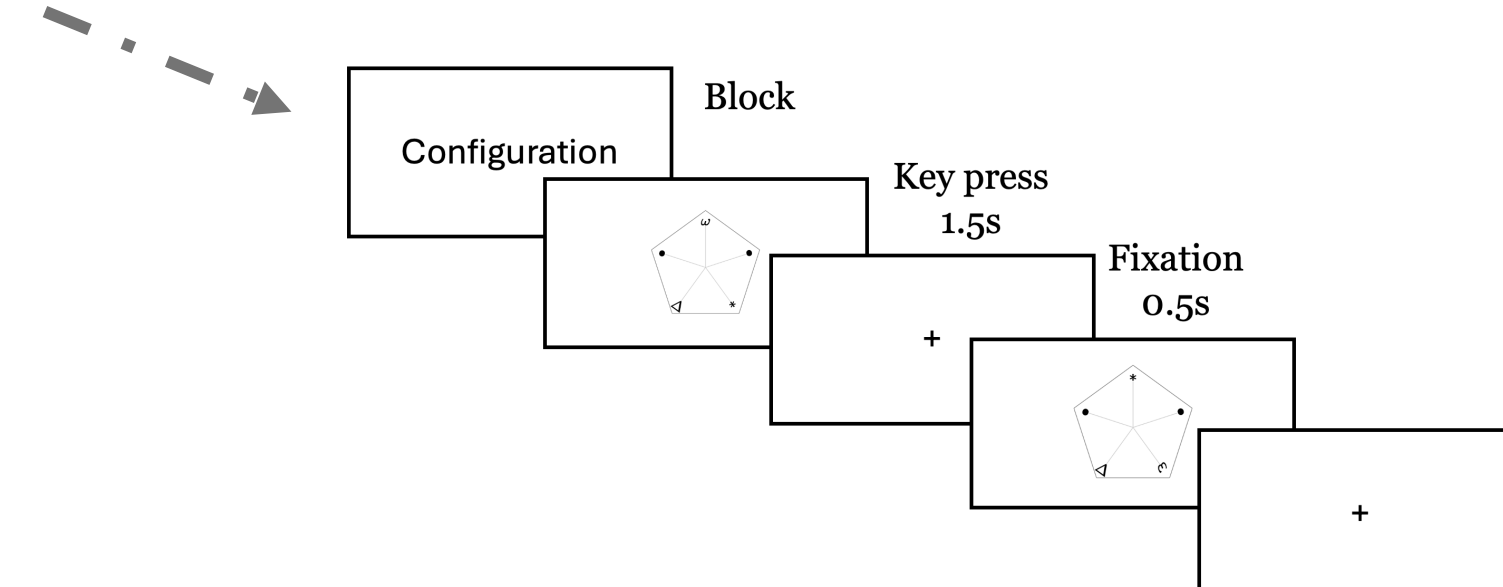
Analysis:

Group-Level Contrast:

- Employed a block design to conduct initial contrasts at the group level between configural and feature processing conditions, identifying overarching patterns of brain activation.

fROI-Based Analysis:

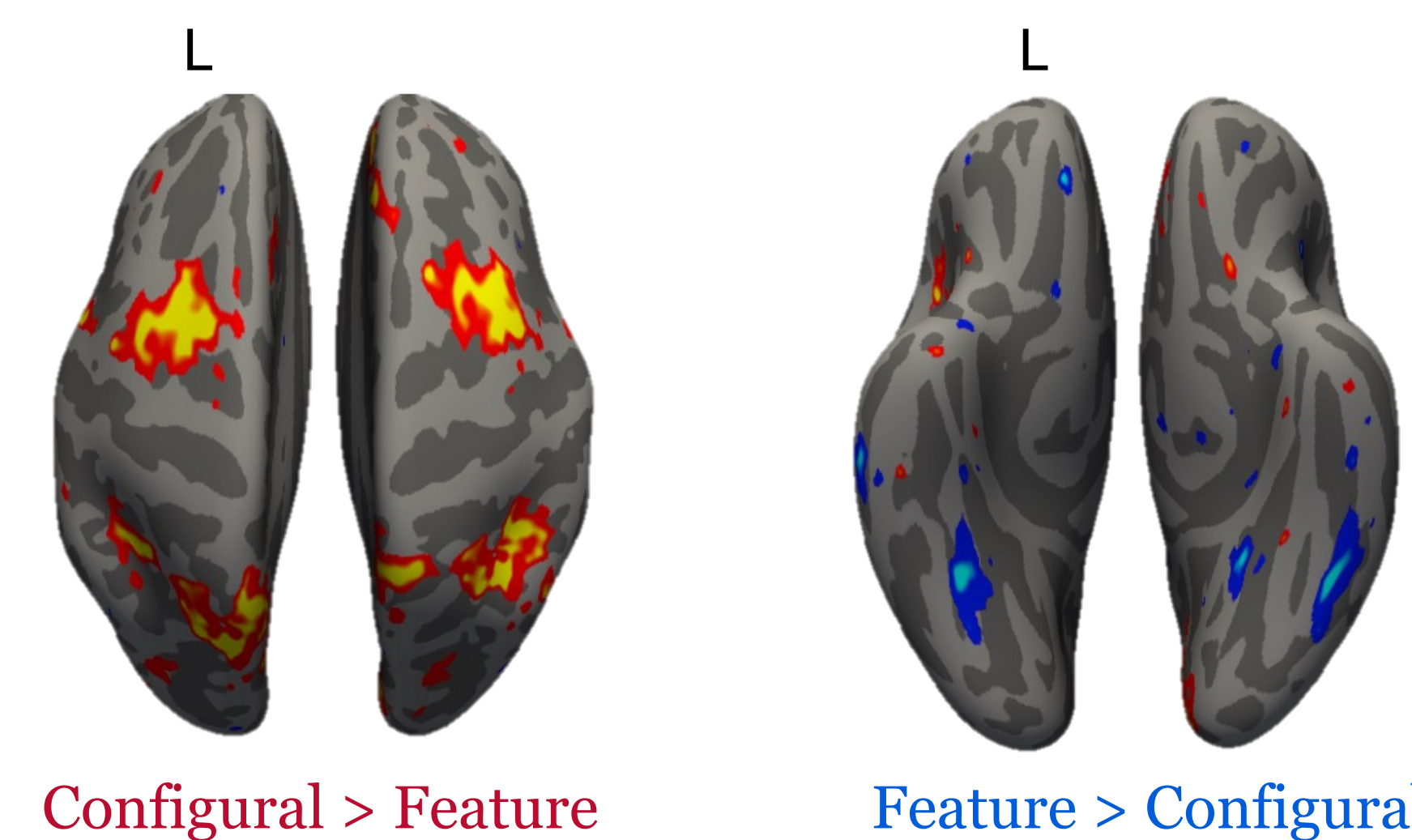
- Utilized the Group-Constrained Subject-Specific (GSS) method [4] to define **individualized functional regions of interest** (fROIs) for each participant.
- Specifically targeted face perception-related areas: Fusiform Face Area (**FFA**), Occipital Face Area (**OFA**), and Superior Temporal Sulcus (**STS**).
- Analyzed percent signal change within these fROIs to assess differences in activation between configural and feature conditions.



Neural Responses to Configural vs. Featural Stimuli

We first performed a **voxel-wise group analysis** to examine the differences in neural activation between configural and feature processing.

Our results indicated that *configural processing* is predominantly localized in the **dorsal areas** of the brain. Conversely, *feature processing* was mainly observed in the **ventral areas**.



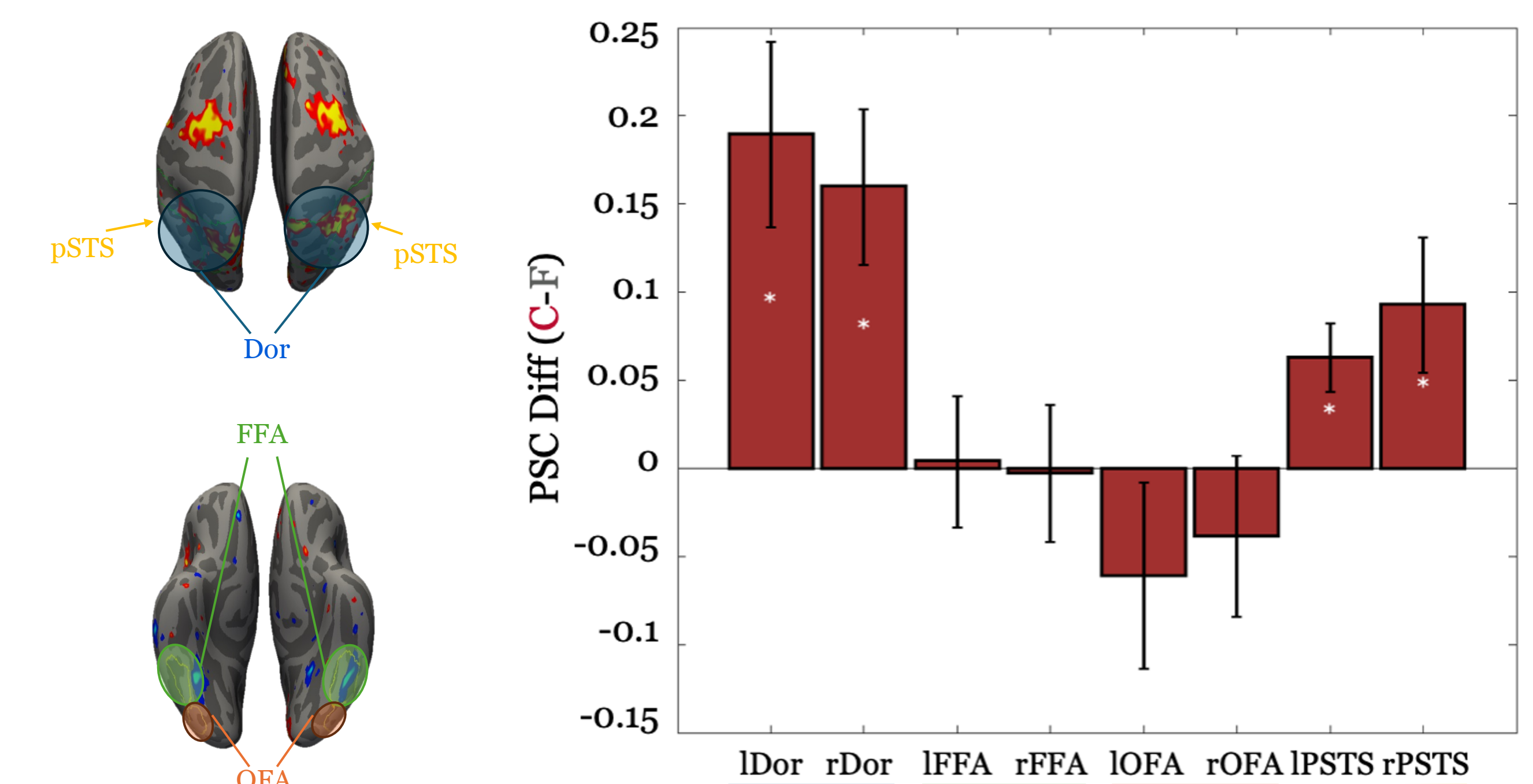
Conclusions

- We found **configural processing** in *non-face stimuli* (and separable from face perception).
- **Configural information** is mostly processed in the **dorsal pathway**, while **feature information** is processed in **ventral pathway**.
- **FFA** have no preference on configural or feature information; **OFA** trends towards feature information; **pSTS** prefers configural information.

Differential Responses in the Face Network

The *dual-pathway model of face processing* delineates the roles of two neural streams: the ventral stream, which is involved in static face form recognition, and the dorsal stream, which manages motion and dynamic face processing [5].

Aligning with this model, our results support the idea that *feature information* is processed primarily through the ventral stream, whereas *configural information* is handled predominantly by the dorsal stream to facilitate **dynamic face processing**.



References

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2. Diamond, R., & Carey, S. (1986). Why faces are and are not special: An effect of expertise. *Journal of Experimental Psychology: General*, 115(2), 107-117.
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5. Bernstein, M., & Yovel, G. (2015). Two neural pathways of face processing: A critical evaluation of current models. *Neuroscience & Biobehavioral Reviews*, 55, 536-546.